FLEET TEAM DIGEST

737MAX-FTD-32-18001

Issue Title : 737MAX PSEU Related Service Interruptions/Delays

Airplane Model	ATA	Minor Model(s)
737MAX	3209-10	-8, -8200, -9, -BBJ

Other Model(s)	ECCN
	9E991

Originated Date	Last Revised Date	Created On
05/03/2018	10/09/2025	10/09/2025

Estimated Completion Date	Next Update Date

Revision Description

Updates were provided to the Milestone Section for closure.

Applicability

All 737MAX Airplanes equipped with the Proximity Switch Electronics Unit (PSEU) PN: 285A1600-101.

Description

In some 737MAX airplanes, a diagnostic and prognostic tool called Airplane Health Management(AHM) is used to integrate remote collection, monitoring, and analysis of airplane data to determine the status of an airplane's current and future service needs. For those airplanes delivered with such service, Boeing is able to take this real-time airplane data into consideration to help airlines make operational and fix/fly decisions. From the customer's perspective, the AHM's fault forwarding capability helps them diagnose and prepare to fix problems before an airplane arrives at the gate.

This Fleet Team Digest (FTD) is written to communicate the following:

1) Acknowledge the ATA 29, 31, 32 and 52 related Flight Deck Effects (FDEs) seen within the delivered 737MAX airplanes

2) To identify the FDEs that are currently being investigated

3) To provide status of the Service Related Problem (SRP) project that was approved in support of the PSEU service interruption/ delay investigation.

To date, Boeing has seen several FDEs that in some cases, have caused service delays while others are just being logged by AHM. Please see attached list of the FDEs that have been extracted from the 737MAX AHM data.

Background

During the March 2018 Fleet Team Meeting, Boeing introduced this topic as a watch item, with focus on the High Level PSEU Flight Deck Effects.

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Boeing has since received reports via Boeing Communication System (BCS) on several observed FDEs (via STATUS or Maintenance Messages) that have caused service delays/interruptions. These reported PSEU faults taken from the 737MAX fleet AHM data are related to ATAs 29 / 31 / 32 / 52. Some are being investigated through a joint effort between Boeing, BAE Systems and Collins Aerospace (CA), while other service reports were corrected by following the existing Aircraft Maintenance Manual (AMM)/ Fault Isolation Manual (FIM) procedures.

The FDE list has been categorized into faults and assigned priority based on number of occurrence and service impact:

1) DOOR SENSOR FAULTS

2) LRU REPLACEMENT TEST FAULTS

3) AIR/GND SENSOR FAULTS

4) CARGO DOOR FAULTS

5) PSEU NO POWER FAULTS

6) PSEU POWER UP FAULTS

7) GEAR LEVER FAULTS

In addition, there are also FDEs that have high occurrences but do not impact the airplane's serviceability and those are categorized as follows:

8) THROTTLE POSITION (TRA) FAULTS AT LANDING

9) TAKEOFF CONFIG HORN FAULTS

To facilitate investigation of these observed FDEs, Boeing has approved an SRP project (737MAX-SRP-52-0137). The Statement of Work of this SRP is as follows:

1) Perform root cause analysis (RCA) on each of the fault categories

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2) Identify all potential solutions

Status

Boeing shares the following results:

1) DOOR SENSOR FAULTS – Investigation complete. Interim action and proposed solution are available under 737MAX-FTD-52-17002.

2) LRU REPLACEMENT TEST FAULTS – Investigation complete. Interim action and proposed solution are available under 737MAX-FTD-32-18003.

3) AIR/GND SENSOR FAULTS – caused by unserviced strut or extra resistance build up on sensor connector. Proposed solution - Revise PSEU SW to:1) Widen acceptable nose gear air/gnd sensor gap range and 2) Tighten physical NLG Air/Gnd sensor rigging gap.

4) CARGO DOOR FAULTS – caused by door switch adjustment tolerance issue. Proposed solution - revise the cargo door switch installation drawing. No PSEU SW fix is required.

5) PSEU NO POWER FAULTS – root cause is unknown. Fault clears when power is restored. Proposed solution - No further action required; fault under monitor.

6) POWER UP FAULTS – 1) Internal Fault -- intermittent low voltage condition (<18V during APU Start from Battery) causes latched faults when voltage returns to normal level. 2) External Door Light Check Fault – intermittent light check failure during power-up. Proposed solution - 1) Revise PSEU SW related to restart routine following low voltage condition to prevent INTERNAL faults. 2) Revise PSEU SW power-up door light test routine to inhibit door faults when door light power is absent or multiple light faults are detected.

7) GEAR LEVER FAULTS – gear lever switch evaluation is ongoing with preliminary results. Microswitch and solenoid failures are suspected to be caused by manufacturing issue. Supplier has implemented risk mitigation steps to resolve manufacturing quality driven issues. BAE/Collins Aerospace/Boeing Team working on improved LRU Screening Test for LG Lever Module and improved Component Screening Test for solenoid. Interim action and proposed solution are available under 737MAX-FTD-32-19001.

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8) THROTTLE POSITION (TRA) FAULTS AT LANDING FAULTS – caused by throttles advanced at landing due to Touch-and-Go, or due to undesired landing conditions.

Proposed Solutions: Revise PSEU SW to change TRA fault conditions to inhibit faults if both throttles are advanced at landing.

9) TAKEOFF CONFIG HORN FAULTS – caused by (1) advancing throttles at gate to sound horn while park brake is set and flaps UP; (2) configuration check done prior to takeoff and while taxiing; (3) advancing throttles during taxi in to auto-stow speedbrake. Proposed solution - No PSEU fix is required; considered normal pilot action.

The PSEU -101 to -102 change affects software only. Airlines may convert PSEU -101 units to the new -102 configuration in accordance with the BAE Systems Service Bulletin 285A1600-32-06, PSEU Upgrade Vendor Service Bulletin. This upgrade requires the PSEU to be removed for software change at the shop level.

Onboard Maintenance Function (OMF) software “46 NFS OMF OSS” provides Ground Test functionality for the PSEU. Due to changes in the -102 PSEU, older 46 NFS OMF OSS part numbers are not compatible with the new -102 PSEU. In order to install and test the -102 PSEU, a compatible 46 NFS OMF OSS must be installed. The -102 PSEU is only compatible with 46 NFS OMF OSS part numbers BCG()()-0MFC-XXXX, where “XXXX” is equal to or greater than 0005. Due to certification requirements, any changes to the 46 NFS OMF OSS will first require upgrading the Onboard Network System (ONS) Operating System (OS) from OS7.1 to OS9.1. A Service Bulletin will be released for all operators by July 2022 to install ONS OS9.1 and includes a newer 46 NFS OMF OSS part number, which supports the -102 PSEU. Reference 737MAX-FTD-46-19001 for additional details.

For airplanes delivered with PSEU -101, the IPC will be revised on 15-May to reflect one-way conditional interchangeability, stating the following – *285A1600-102 CAN REPLACE BUT NOT BE REPLACED BY 285A1600-101. INSTALLATION OF 285A1600-102 REQUIRES 46 NFS OMF OSS SOFTWARE PART NUMBER BCG3B-0MFC-0005 (ATN BCG3B0MFC0005) OR LATER VERSION*.

Interim Action

Reference Interim Actions in the following:

737MAX-FTD-52-17002

737MAX-FTD-32-18003

737MAX-FTD-32-19001

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Final Action

Final Action for the (5) fault categories are as follows:

1) DOOR SENSOR FAULTS – Proposed solution is available under 737MAX-FTD-52-17002. MAR-2019 (complete)

2) LRU REPLACEMENT TEST FAULTS – Proposed solution is available under 737MAX-FTD-32-18003. MAR-2019 (complete)

3) AIR/GND SENSOR FAULTS – 1) Revise PSEU SW to widen acceptable nose gear air/gnd sensor gap range. 2) Revise NLG Sensor Installation Drawing to tighten physical NLG Air/Gnd sensor rigging gap. MAR-2019 (complete)

4) POWER UP FAULTS – 1) Internal Fault - Revise PSEU SW related to restart routine following low voltage condition to prevent INTERNAL faults. 2) Identify PSEU fix for Door Light Check failure. Fix to be included in PSEU -102 SW update. MAR-2020 (complete)

5) THROTTLE POSITION (TRA) FAULTS AT LANDING FAULTS – Revise PSEU software to change fault set conditions for 6 TRA faults at landing and to only declare fault if a single left or right TRA signal is in incorrect state 5 seconds after landing. MAR-2019 (complete)

The Final Action would be for operators to incorpate SB 737-32-1555 to upgrade the -101 to -102 PSEU configuration.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Identification: MAY-2018 (complete)

Root Cause Established: DEC-2018 (complete)

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Root Cause Approved: DEC-2018 (complete)

Solution Approval: AUG-2019 (complete)

Production Incorporation -102 PSEU SW and MDS for (737-7/-8/-9): OCT-2020 (complete)

Supplier Service Bulletin Available (737-7/-8/-9): SEPT-2020 (complete)

Boeing Service Bulletin (737-32-1555): MAY-2022 (target) (complete)

Boeing Service Letter: APR-2025 (complete)

Operator Action

See Interim Action.

Supplier Information

For matters related to the PSEU:

BAE Systems

Fort Wayne Service Center

2000 Taylor St. Dock 1

Fort Wayne, Indiana 46802-4605 USA

E-mail: baefwaservctr@baesystems.com

Phone: 260-434-5600

Fax: 260-434-5616

BAE Systems Website: http://www.baesystems-ps.com/customersupport/

For matters related to the Proximity Sensors:

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Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737MAX-FTD-46-19001		
Fleet Team Digest Article (FTD)	737MAX-FTD-32-19001		
Fleet Team Digest Article (FTD)	737MAX-FTD-32-18003		
Fleet Team Digest Article (FTD)	737MAX-FTD-52-17002		
Service Related Problem (SRP)	737MAX-SRP-52-0137		
Service Letter (SL)	737-SL-32-238		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Proximity Sensing Electronics Unit	285A1600-101	BAE Systems	

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- AOGs : Events that cause a significant Airplane On Ground (AOG) condition
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- SRPs - Economic : Service Related Problems in work by Boeing that have an economic impact

Attachments:

a) Attachment I - 737MAX PSEU Related FDEs

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